Supplementary material for 'A generative rule from affect to tactile relief: what it decides, what its spectral descriptors measure, and what classification accuracy checks'

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S1 Generator and descriptors

Source: experiments/spec/generator-and-descriptors.md (version 1, SHA-256 0ab26540..., committed in fdb9b71 before the computations of the paper). The specification was written before those computations; if the code and the document disagree, the document is the specification. The clauses are summarized here.

Field. The domain is the unit square, sampled at $x_i = x_0 + i/N$, $y_j = y_0 + j/N$ ($i, j = 0, \ldots, N-1$); the default window is $(x_0, y_0) = (0, 0)$, and shifting it gives another window on the same infinite field. The field is that of Section 2 of the main text, with four octaves. The phases are drawn by `numpy.random.default_rng(seed)`, two per octave in the order ϕ_o, ψ_o, uniform on $[0, 2\pi)$; no other draws occur. The canonical normalization is $F = F_{\mathrm{raw}} / \max |F_{\mathrm{raw}}|$ over the grid ('maxabs'); the variant 'ampsum' divides by the constant $\sum_o 2^{-o}$, so that every window and grid samples the same surface, and is used where windows or grids of one surface are compared. The centroid does not depend on this choice. The canonical generator has no domain warp; the warp is a variant used only in Section S7. F is periodic on the domain if and only if f is an integer.

Height and descriptors. (f, s, γ) follow from (V, A) by the rule of Section 2 of the main text, with V and A clipped to $[-1, 1]$. $H = s\, T_\gamma(F)$, with the smoothed variant $T_{\gamma,\varepsilon}$; descriptors are computed on $H - \overline{H}$. The amplitude-weighted centroid sums over all N^2 bins, each component of k taking the integers in $[-N/2, N/2)$, with $\rho_k = |k|$ in cycles per domain. The power-weighted and band-fixed centroids replace $|\hat{H}|$ by $|\hat{H}|^2$ and restrict the sums to $0 < \rho_k \le K$, with K in cycles per domain and $K \le N/2$. S_q is the root mean square of $H - \overline{H}$. The maximum gradient uses `numpy.gradient(H, 1/N)` (central differences inside, one-sided at the edge). The normal map of a height uses the same differences with spacing L/N, the OpenGL convention and uint8 encoding. The high-frequency ratio of Gao et al. (2026) is copied from the authors' public code; its cutoff is 120 pixels ('px') or $120N/N_{\mathrm{ref}}$, fixed in cycles per domain ('phys').

Edge treatments. Window $W(i, j) = \sin^2(\pi i/N) \sin^2(\pi j/N)$ applied to $H - \overline{H}$, then the mean removed; the periodic component of Moisan (2011); and the band-fixed centroid. None gives a 'true' value.

Aliasing. A reference is made by generating H on a grid of side $2N$ (and $4N$ for $N \le 2048$ as a check), keeping the bins in $[-N/2, N/2)$ and transforming back; plain subsampling is the control that keeps aliasing.

Measured data. Surface-Topography Challenge (Zenodo, doi:10.5281/zenodo.15341939, CC0): the variable `heights` of each digital surface twin, converted to metres, cropped to the central $P \times P$ block (P the largest power of two not above the shorter side), with the least-squares plane removed; coarser levels $P/2$ and $P/4$ by the Fourier reduction of S4 and, as a control, by subsampling. TactileDreamFusion (Hugging Face, revision 87c8b764…, MIT): the normal patches (240×240) and the central 1024×1024 block of the tiled maps, reduced per component and re-encoded.

Statistics. A growth exponent is the least-squares slope of $\ln C$ on $\ln N$ per seed, summarized by the mean and the 95% t interval over seeds, with Bonferroni intervals when a decision is taken over several cells. A flip of the order of two surfaces between two grid levels is a pair whose sign differs and whose mean difference exceeds twice its standard error at both levels; for measured data without seeds a relative difference above 0.05 at both levels is used. Package versions are fixed by `uv.lock`, and every results file records the SHA-256 of its conditions file and of the specification.

Table S1: Clauses of the specification, the code that implements them and the tests that check them.

Clause	Code (`src/atact/`)	Test (`tests/`)
S1.1–S1.4	`field.procedural_field`, `RenderConfig`	`test_spec.py::test_s1_*`
S1.6	–	`test_spec.py::test_s1_6_periodic_iff_integer_f`
S2.1–S2.2	`mapping.params`, `field.height`, `transform`	`test_spec.py::test_s2_2_*`, `test_math.py`
S2.3–S2.7	`descriptors`, `spectral.centroid_band`	`test_spec.py::test_s2_*`
S2.8–S2.9	`hfratio`, `spectral.linear_hf_ratio`	`test_hfratio.py`, `test_spec.py::test_s2_9_*`
S3	`spectral`	`test_spec.py::test_s3_*`
S4	`spectral.decimate_fourier`, `subsample`	`test_spec.py::test_s4_*`
S5	`spectral.remove_plane`	`test_spec.py::test_s5_*`

S2 Conditions fixed before each computation

Source: experiments/conditions/INDEX.json. Each computation reads a conditions file that was committed to the repository before the computation was run, and records the SHA-256 of that file and of the specification in its results. The commits are those of the repository; they are not externally time-stamped. A change after a file was committed raises its version and is listed under the file's deviations and in the history of the index.

The observable degrees of freedom (Section S6) come from an earlier computation (conditions observable_dof.json, SHA-256 63cd7758…, results 0f320bc), whose settings agree with the specification, and were not recomputed.

Deviations recorded in the history of the index: edge_grid_paired went to version 2 before its run because version 1 stated wrongly that the rule points with $A \neq 0$ have non-integer f (five points with $f = 5.5$ were added), and to version 3 before its run because a control could not produce the flip it was meant to show. text_path_example went to version 2 after its run, to redraw the figures from the stored results without changing any value. The jitter stream of K5 was changed before any K5 computation (Section S3).

S3 Statistical procedures

Source: experiments/conditions/k5_classification.json (version 1) and the reading of the conditions in results/k5_classification.md, fixed before the run.

Table S2: Conditions files, the commit that added each (before its computation) and the commit of the results.

File	Version	SHA-256	What it decides	Conditions	Results
k4_centroid_2d	1	376a638c…	K4 (b)–(d), Appendix B	ac1c522	cb369b4
edge_grid_paired	3	1ca8bfcb…	the edge and the grid on the rule	ac1c522	32e2ffe
validation3_public	1	17d233cb…	measured surfaces	ac1c522	0acb76d
k5_classification	1	09a91416…	K5 (Table 1)	ac1c522	3f58861
k5_table1_predictions	1	6cf0f969…	the twelve predictions of K5	2de1d67	3f58861
variants	1	24516ef8…	Section S7	ac1c522	e41568a
k2_gradient_and_figures	1	1f1570a3…	K2, the example of ε	c43c855	59c213e
text_path_example	2	c453a2b2…	the text example	c2a325e	ad02721
k4_supplement*	1	790ad9d1…	K4 up to $N = 16384$	b7f9911	cadd349
v3_reduction_supplement*	1	7f2f6108…	reduction methods, measured data	693c234	40bf399
cband_cutoff_sensitivity*	1	ff062405…	C_{band} and the band limit	8979816	e168ba7

* Exploratory: designed after the results it extends had been seen; not evidence for the claims of the main text (Section S5).

Groups and jitter. Group 0 is the first centre of each condition and group 1 the second. Surface g with seed s draws u uniform on $[-1, 1]^2$ from `default_rng([s, g, 1])` and is jittered by wu. (The decision before any computation named `default_rng([s, g])`; a check before the power runs found that this repeats the first phase draws of the field for group 0, and the third word was added.) The conditions with s or s and γ fixed follow the second author's validation materials, which separate the mechanisms by fixing parameters. Condition (i) with V varied adds wu_1 to V only, and with A varied wu_2 to A only; the conditions at the corners add both and clip to $[-1, 1]$ (0.758 of the surfaces have a clipped coordinate, at every w). The same u is used for the three values of w. Pair i of the experiment uses seed $1000 + i$ for both groups, except the negative control, whose group 1 uses $50000 + i$.

Cross-validation and classifiers. Five folds by `GroupKFold`, with both surfaces of a pair always in the same fold and the folds fixed by the pair indices. The SVM is a linear support vector classifier ($C = 1$) after standardization, fitted on the training folds. The threshold is the cut and direction that maximize the training accuracy. Accuracy is the fraction of the $2n$ surfaces predicted correctly out of fold.

Intervals and tests. The pairs are resampled with replacement (2000 replicates, the same for every row) with the out-of-fold predictions kept fixed, so the percentile interval does not include the variation from refitting the classifier. 'At chance' means that the 90% interval of the accuracy minus 0.5 lies inside $(-0.05, 0.05)$ (two one-sided tests). A permutation test (999 replicates, the labels of each pair swapped with probability 1/2, the whole pipeline rerun) gives a p value for each classifier. Because the interval omits the variation from refitting, it is too narrow, so equivalence is declared more easily than its nominal level suggests.

Sample size. Candidates of 60, 120, 240 and 480 pairs were run on independent seeds before the experiment: (a) 200 runs of the negative control, requiring that equivalence be declared in at least 80% of runs in every column; (b) 50 runs at the corners, requiring a mean half-width of the 95% interval of at most 0.05. No candidate met (a); the smallest per-column rate at 480 pairs was 0.73 (results/k5_power.md), and the largest candidate was used, as the rule stated. The power runs predicted on average 12.3 columns of the negative control not declared at chance; the experiment had 11.

The level. Two further samples of 5000 surfaces per group (seeds from 2 000 000 and 3 000 000) give the level. For one descriptor it is the accuracy of the best threshold on the first sample, which is biased upwards by the selection; its accuracy on the second sample is reported beside it in results/k5_large_sample.md. For two descriptors it is the SVM of the experiment trained on the first sample and tested on the second. For the conditions where the threshold and the SVM differ (Section S4, prediction P-K5-12) the level of one descriptor does not predict the SVM. In the frame of Table 1 written before the experiment, the level of one descriptor was entered for both classifiers; the main text and Table 1 state it as a level for the threshold only. On the second sample the accuracy of the threshold differs from the level by at most 0.008 outside the negative control and 0.013 within it (the largest differences over the rows of results/k5_large_sample.md).

S4 Full results

The results files named below are part of the supplementary fileset. Each gives the predictions as committed before the run, the values, and conclusions written by hand after the generated results had been committed unedited. This section summarizes them and gives the details that the main text refers to.

Table S3: Registered computations and the outcome of their predictions (a prediction 'holds' only if it holds in every item).

Computation (results file)	Predictions holding in every item	Not in every item
K2 (k2_gradient.md)	P-G1–G4, G6	P-G5 (1 of 3: only for the main ε)
K4 (k4_centroid_2d.md)	P-K4-1, 3, 5	P-K4-2 (9/12), 4 (0/6), 6 (25/30), 7 (61/66)
Edge and grid (edge_grid_paired.md)	P-E2, 4, 6, 7	P-E1 (4/5), E3 (0/1), E5 (3/4)
Measured data (validation3_public.md)	P-V2, 4, 5	P-V1 (0/2), V3 (1/2)
K5 (k5_classification.md)	P-K5-4, 5, 6, 9, 11	P-K5-2 (5/6), 3 (1/2); 1, 7, 8, 10, 12
Warp and Table 1 (variants_reported.md)	—	the warp leaves the K5 results unchanged (Section S7)

K2. Over 20 seeds and $N = 512$–4096 the slope of $\ln \max |\nabla H|$ on $\ln N$ is 0.425 $[0.412, 0.437]$ at $\gamma = 1.75$ and 0.606 $[0.592, 0.619]$ at $\gamma = 2.6$ (95% t intervals over seeds), against $1 - 1/\gamma = 0.429$ and 0.615, and 0.003 at $\gamma = 0.9$. For a single sine the closed form of σ_γ predicts a ratio of 5.98 in S_q between the corners $(V, A) = (-1, 1)$ and $(1, -1)$, a factor 5 of it from s alone. The slopes of the largest gradient were computed with the field divided by the constant $\sum_o 2^{-o}$ ('ampsum') instead of its maximum, so that every grid samples the same surface; the canonical maxabs normalization gives the same slopes to the third decimal. P-G5 required the smoothed slope to stop changing for three values of ε; it holds for the main value 0.353 (1.9% between $N = 1024$ and 4096) and not for the two smaller ones. At the steepest point of the rule ($f = 7$, $s = 0.30$, $\gamma = 2.6$, a tile of 50 mm on the common scale) the steepest slope of T_γ rises from 59° at $N = 128$ to 86° at $N = 4096$ (means over 20 seeds); with $T_{\gamma,\varepsilon}$, $\varepsilon = 0.353$, it is 61° at $N = 4096$, and the height changes by at most 0.13 mm on a relief that spans 1.02–1.11 mm. With $\varepsilon = 0.044$ (a band of 0.05 mm) the slope still grows by a factor 1.38 between $N = 1024$ and 4096: it stops changing only once the grid resolves the band.

K4. P-K4-2 fails at $\gamma = 0.9$, where the slopes of one octave of the product field are 0.055–0.063, neither close to the negative target $1 - 1/\gamma$ as written nor within 0.05 of 0; their local slopes decrease with N, as in one dimension. P-K4-4 fails as operationalized: at $\gamma = 1$ and integer f the centroid is constant to a relative range of 9.1×10^{-11}, but floating-point rounding makes the slope intervals exclude 0. P-K4-6 fails in five items at $\gamma \geq 1.75$: in three the windowed slope for non-integer f is not smaller than the raw slope, and in two it does not approach the windowed slope for integer f. Over $N = 512$–4096 at $\gamma = 1.75$ the one-dimensional slope is 0.425–0.426

(against 0.429) and the slope on one octave of the product field is 0.453–0.461. The intervals over seeds do not include the bias of a finite range of N (Appendix B of the main text). With integer f the power-weighted centroid of the sum field stops changing with N (local slopes at most 0.007 between $N = 2048$ and 4096); with non-integer f it keeps growing, by about 0.3–0.4 per unit of $\ln N$ at the largest N. On the sum field the slope of $\ln C$ on $\ln N$ exceeds the one-dimensional exponent in all 12 cells ($f \in \{3, 5, 7\}$, four values of γ; P-K4-3), with 0.287–0.304 at $\gamma = 0.9$ and 0.902–0.905 at $\gamma = 2.6$ against 0.615 in one dimension. At $\gamma = 1$ (means over 20 seeds) C rises from 20.8 at $N = 128$ to 187.1 at $N = 2048$ and 349.7 at $N = 4096$ for $f = 5.5$ and stays at 15.08 for $f = 5$; the slope over $N = 512$–4096 is 0.82–0.89 for $f = 3.5$, 5.5 and 6.3. The raw slopes for non-integer f (3.5, 5.5, 6.3) against integer f (k4_centroid_2d.md, P-K4-6) are, on one octave of the product field, 0.874–0.888 against 0.059 ($\gamma = 0.9$), 0.876–0.889 against 0.232 ($\gamma = 1.25$), 0.849–0.882 against 0.457 ($\gamma = 1.75$) and 0.811–0.863 against 0.629 ($\gamma = 2.6$); on the sum field, 0.813–0.876 against 0.295, 0.798–0.858 against 0.619, 0.795–0.840 against 0.811 and 0.843–0.883 against 0.903. The edge raises the slope at every γ on the product field, and on the sum field only at $\gamma \leq 1.25$ over this range of N.

The window position. At $N = 2048$ and $\gamma = 1$, shifting the window on the infinite field changes C by 8.5–33.7 at five points with $f = 5.5$ and by 0.13–0.20 at the corresponding points with $f = 5$. This is 2–8% of the span of C over the 5×5 points of the rule at the same N (431). It measures how much the centroid depends on where the window is cut, not how much the edge adds to it, which grows with N (Section 4(c) of the main text). The span itself depends on the grid: between $N = 128$ and 2048 the centroid of a point changes on average by 0.49 of it. A window, the periodic component of Moisan (2011) and the band-fixed power centroid each reduce the dependence on the position of the window at the five points with $f = 5.5$, at every N (P-E4: 25 of 25 cells for each; edge_grid_paired.md).

Measured data. This tests robustness to resampling, not to re-measurement, and says nothing about the studies that use these descriptors. The 38 height measurements are those of the Surface-Topography Challenge with a shorter side of at least 512 points and no undefined data (33 of the rougher and 5 of the smoother wafer). A flip is counted for pairs from different containers over the three pairs of grid levels, with a relative difference above 0.05 at both levels; the 38 measurements give at most 703 pairs, so at most 2109 comparisons, and the 33 rougher and 5 smoother measurements give at most 495 comparisons of a rougher with a smoother one. On the normal maps every pair of the 36 inputs is counted (630 pairs, 1890 comparisons). The high-frequency ratio with the cutoff in pixels has 126 flips and with the cutoff in cycles per domain 61; on the 36 normal maps the counts are 118 and 0. The amplitude-weighted centroid has 100 flips, 32 of which reverse the order of a measurement of the rougher and one of the smoother wafer; the two ratios reverse no such pair. P-V1 fails in both of its items; in the first, the pixel ratio increases strictly with coarsening for 35 of 38 measurements, and two of the exceptions are smooth measurements whose ratio is already 1 at the coarser levels.

The twelve predictions of K5. Written from K1–K4 before any K5 computation (k5_table1_predictions.json). The predictions that apply to each cell of Table 1 of the main text, and whether they hold there, are listed in results/table1_main.md. The slice shown in Table 1 ($N = 128$, no window, $w = 0.3$) was fixed in the extraction script after the 544 rows had been committed (commits 3f58861 and 39ef4bc); $N = 128$ is the grid of the preprint. Whether a prediction holds is decided on the point estimates; differences smaller than the width of the intervals (about 0.01–0.03 in Table 1) cannot be told apart. Predictions 3 and 12 concern the level (at chance or above it); the other ten concern direction and order. A prediction may have several parts (separate statements) and is checked in every cell, row or column it applies to; it holds 'in every part' only if every part holds in every one of them.

1. S_q on the min–max export is at most S_q on H at the corners. Fails in 4 of 36 cells, all with both values at chance.
2. For (i) with A varied, C does not fall from $N = 128$ to 512, and C_{band} and the windowed C change less. 5 of 6 parts hold (one surface out of 960 against).
3. The negative control is at chance in every column and the same-seed control is exactly 0.5. The second part holds; the first fails in 11 of 56 columns. The negative control lies between 0.461 and 0.514 (mean 0.483 over the 56 columns), with permutation p values from 0.27 to 0.98; the mean over the power runs at this sample size was 0.4996, so the shift below 0.5 is one realization on the low side.
4. C and C_{band} are the same on H and on the min–max export, with equal accuracies. Holds.
5. The condition with s fixed and the rule at the corners agree after the min–max export. Holds.
6. For (i) with V varied, S_q is larger at the smaller V in every pair. Holds in all 2880 pairs.
7. The SVM on (S_q, C) is at least as accurate as the better of the two. Fails in 7 of 120 cells, by at most three surfaces.
8. Accuracy does not increase with w. Fails in 10 of 160 columns (six at chance, four by one surface).
9. S_q with s and γ fixed is below S_q in every other condition. Holds; its stated basis was partly wrong: for the integer $f = 3$ or 7 of the clipped inputs, coprime to N, the samples of the two surfaces of a pair are permutations of each other and their S_q is identical, and elsewhere it differs only through the edge of the domain.
10. The centroids are at least as accurate as S_q where f differs, and S_q at least as accurate as C for (i) with V varied. Fails in 3 of 96 cells, by two surfaces.
11. For (i) with V varied, C does not fall from $N = 128$ to 512. Holds.
12. Every non-null row is above chance except S_q with s and γ fixed. Fails in 12 of 456 rows: the SVM on C and on (S_q, C) with s and γ fixed at $N = 512$ (accuracy 0.506–0.519, against 0.761–0.773 for the threshold). A reading, not a tested mechanism: at $w = 0.1$, $N = 512$, the centroid of each group has a point mass at its minimum (9.05 and 21.12, also the medians) (the clipped inputs, whose f is the integer at the corner) and a long upper tail (maxima 63.9 and 72.2); the hinge and logistic losses are pulled by the tail of group 0, and their cut does not fall between the two masses, while the best threshold does. At $N = 128$ the tails are short and the SVM reaches 0.90 on all data (results/k5_classification.md).

The full table of the 544 rows is in results/k5_classification.md.

S5 Exploratory computations

The three computations of this section were designed after the results they extend had been seen. They are not blind and are not evidence for the claims of the main text; the main text uses them only to state limits.

K4 up to $N = 16384$. Source: results/k4_supplement.md (commit cadd349). The values for $N = 128$–4096 agree with K4 to a relative difference of 6.7×10^{-14}. In all 12 cells the local slopes of the sum field keep increasing up to $N = 16384$ (for example $f = 5$, $\gamma = 0.9$: 0.312, 0.327, 0.339 over the last three doublings; $\gamma = 2.6$: 0.918, 0.929, 0.938), with smaller increases per doubling from $N = 512$ on. Over $N = 4096$–16384 every interval still lies below the heuristic exponent $e(\gamma)$ (for $f = 5$: 0.333 against 0.389, 0.665 against 0.700, 0.847 against 0.929, 0.934

against 1). Three extrapolations of the local slopes lie on both sides of $e(\gamma)$, depending on their form. The exponent as $N \to \infty$ remains open. The dotted points of Fig. 4(c) of the main text come from this computation.

Reduction methods for measured data. Source: results/v3_reduction_supplement.md (commit 40bf399). The counts of flips do not depend on the reduction method in the direction that matters: the pixel ratio has 126, 133 and 143 flips under the Fourier reduction, mirroring and subsampling, and the ratio with the cutoff in cycles per domain fewer under each (61, 84, 71); the amplitude-weighted centroid reverses a rougher and a smoother measurement 32, 29 and 25 times, the ratios never.

The band limit of C_{band}. Source: results/cband_cutoff_sensitivity.md (conditions cband_cutoff_sensitivity.json, version 1; commits 8979816, e168ba7, b89078a). $K = 32$ lies inside the range of the rule: the octave peaks of the field lie at radius $\sqrt{2}\,2^o f$, so with A varied octave 2 of group 1 crosses $K = 32$ at $f = 5.66$. The K5 accuracies of C_{band} were recomputed for $K = 16$, 24, 32, 40, 48 and 64 on the surfaces of the K5 experiment (H, both grids, both processings, all w); at $K = 32$ they equal the stored K5 accuracies in all 120 classifier rows. With A varied the accuracy changes from its value at $K = 32$ by at most 0.002 at $w = 0.1$ and 0.3 and by at most 0.018 at $w = 0.5$. With V varied ($f = 5$, octave peaks at radius 7.1, 14.1, 28.3 and 56.6) the accuracy lies between 0.554 and 0.887 and is not monotone in K: lowest at $K = 16$, highest at $K = 24$, which excludes octave 2, and 0.68–0.70 at $K = 32$, which includes it. At the corners and with s fixed, alone or with γ, it does not change with K. The level of a band-fixed descriptor therefore depends on where the band ends relative to the octaves; the main text states this as a limit of the band-fixed centroid (Section 5 of the main text).

Printed lengths. Computed by hand from the definitions, not by the scripts. On a 50 mm tile with $f = 7$ the finest octave has 56 cycles along each axis, a period of about 0.9 mm. Keeping only features of 2 mm takes a band limit of $50/2 = 25$ cycles per domain. The octave peaks lie at radius $\sqrt{2}\,2^o f$ with $3 \le f \le 7$, whose union covers about 4 to 79 cycles per domain without a gap, so every limit in that range cuts some octave for some inputs; a higher limit keeps all octaves, with the harmonics of T_γ and the leakage. The ε of Section 3 of the main text is tied to a 0.4 mm nozzle; by the same relation a band of 2 mm needs ε of about 1.8 at the steepest point, beyond the range where smoothing is a small change of shape (the bound $\varepsilon^{1/\gamma}$ of Proposition A.3 of the main text exceeds 1). How much of the shape set by γ remains there was not computed.

S6 Observable degrees of freedom

Source: results/observable_dof.md (conditions observable_dof.json, commit 0f320bc). The rank of the Jacobian of the map from the inputs to each stage was computed numerically by central differences at four points $(V, A, D) = (0, 0, 0)$, $(0.5, -0.5, 0)$, $(-0.5, 0.5, 0)$ and $(0.3, 0.2, 0.5)$ with seeds 1, 2 and 3, at $N = 128$, with the canonical generator. The rank is the number of singular values above 10^{-6} times the largest. Besides the rule, a hypothetical third input $D \in [-1, 1]$ adds $0.3D$ to γ and leaves f and s unchanged; it is the smallest change that moves γ alone, not a proposal for the rule.

With three inputs the min–max export has the null direction $(dV, dA, dD) = (1, 0, 0.992)$ (scaled by its largest component): a change of V that moves s and γ is undone by a change of D that restores γ, and the export removes s. The descriptors give rank 2 because there are two of them.

Table S4: Numerical rank of the map from the inputs to each stage (the same in all 12 combinations of point and seed), and the range of the smallest relative singular value.

Inputs	Stage	Rank	Smallest relative singular value
(V, A)	H	2	1.7×10^{-2}–6.1×10^{-2}
(V, A)	min–max export	2	2.1×10^{-3}–7.6×10^{-3}
(V, A)	(S_q, C) on H	2	2.3×10^{-4}–7.0×10^{-3}
(V, A)	(S_q, C) on the export	2	4.8×10^{-5}–1.0×10^{-2}
(V, A, D)	H	3	1.1×10^{-3}–4.1×10^{-3}
(V, A, D)	min–max export	2	3.2×10^{-14}–9.7×10^{-14}
(V, A, D)	(S_q, C) on H	2	2.3×10^{-4}–8.9×10^{-3}
(V, A, D)	(S_q, C) on the export	2	6.8×10^{-5}–1.3×10^{-2}

S7 The reported accuracies and the variants

Source: results/variants_reported.md (conditions experiments/conditions/variants.json, version 1; run at commit c5e691c; conclusions written after the run). The preprint reports 91.6% for R_q (S_q here) alone and 74.2% for the centroid alone at the corners $(1, -1)$ and $(-1, 1)$, 60 surfaces each. Its procedure leaves open the noise level, the jitter width w, the domain warp, the grid N, the window and the definition of the centroid. The conditions fix a full factorial of these, $3 \times 32 \times 3 \times 2 \times 2 \times 2 = 2304$ variants, each run on 20 replicates of the data set. For each variant and each column, Δ is the accuracy minus the reported value, with the percentile interval over the replicates; a variant can explain a reported value if this interval contains 0. The list of variants was fixed after exploratory computations had been seen. The comparison of the warp with Table 1 below was fixed before it was run, and its expectation failed (it is listed with the failed predictions in Section S4). Nothing in this section is evidence for the claims of the main text; the main text uses it only to state limits.

Reported values. The interval contains 0 for S_q alone in 24 of the 72 combinations that S_q depends on, for the centroid alone in 160 variants, and for both in 101, all of which also reproduce the reported order (S_q above the centroid). All 101 use the warp (0.25) and $w = 1.25$ or 1.5; they cover all noise levels, both grids, both window treatments and 28 of the 32 centroid definitions. The accuracy for S_q alone is set by w: its mean is 0.944–0.952 at $w = 1.25$ and 0.873–0.878 at $w = 1.5$ without the warp, and 0.940–0.948 and 0.874–0.879 with it. That no combination without the warp contains 0 is a margin of 0.002–0.009 at the end of an interval from 20 replicates, not a difference in the mean. The comparison therefore does not identify the original procedure.

The warp and Table 1. All conditions of K5 at $N = 128$, without a window and on H, were computed with warp 0.25 and compared with warp 0. As an implementation control, the warp-0 accuracies equal the stored K5 accuracies in all 119 rows. The conditions describe this computation as a check that the warp does not change the K5 results; it does change them. The largest difference is 0.347: under (i) with V varied the centroid falls from 0.93–0.99 to 0.64–0.66 at every w, under (i) with A varied S_q falls from 0.88–0.96 to 0.70–0.74, and under (ii) with s and γ fixed the centroid rises from 0.84–0.96 to 0.98–1.00. The order of the descriptor sets changes in some rows, for example C below C_{band} at warp 0 and above it at warp 0.25 under (i) with A varied. With the rule at the corners and with s fixed the accuracies change by at most 0.043, and the controls stay near 0.5. With the warp, the generator draws the warp layers before the octave phases, so each difference also contains a redraw of the surfaces; no interval of the difference was computed. Whether the original generator used a warp, and with what amplitude, is not known. The levels and the order in the main text are those of the field without warp.

S8 The text example

The main text shows one text (B). Here text A, 'Is this the place? Where am I? Who am I?', is shown beside it, with all matched terms (Fig. S1) and with function words removed (Fig. S2). Source: results/text_path_example.md (conditions text_path_example.json; results ad02721). Seed 42, $L = 50\,\text{mm}$, $D = 1.2\,\text{mm}$, $s_{\max} = 0.30$, $\varepsilon = 0.353$. Text B is one expansion of text A, not a sample of texts. Two sentences of text A match terms ('where', 'am') whose valence and arousal are 0 in the lexicon, and so land exactly at $(0, 0)$, where they cannot be told from a sentence that matches nothing; without function words they match nothing.

Table S5: Both texts through the rule. Z: matched terms. Span: relief on the common scale (mm). Angles: steepest slope in degrees at $N = 128$ / 1024 on the common scale, under min–max, and on the common scale with $T_{\gamma,\varepsilon}$.

	Sentence	Z	V	A	f	s	γ	common	min–max	$T_{\gamma,\varepsilon}$
All matched terms										
A1	Is this the place?	3	0.049	−0.159	4.681	0.174	1.648	30.8 / 51.5	51.0 / 69.0	22.9 / 26.7
A2	Where am I?	2	0.000	0.000	5.000	0.180	1.750	34.5 / 57.0	53.7 / 71.8	25.5 / 30.0
A3	Who am I?	1	0.000	0.000	5.000	0.180	1.750	34.5 / 57.0	53.7 / 71.8	25.5 / 30.0
B1	Is this the old station …	5	0.034	−0.083	4.834	0.176	1.694	32.1 / 53.8	52.0 / 70.2	24.3 / 28.1
B2	Where am I, lost …	7	−0.142	−0.132	4.736	0.197	1.719	34.7 / 58.1	51.7 / 71.1	24.9 / 30.6
B3	Who am I, when no one …	5	−0.014	−0.268	4.465	0.182	1.606	30.4 / 51.0	49.4 / 67.9	22.6 / 26.3
Function words removed										
A1	Is this the place?	1	0.146	−0.478	4.044	0.163	1.443	23.0 / 38.9	44.0 / 61.4	17.4 / 20.4
A2	Where am I?	0	0.000	0.000	5.000	0.180	1.750	34.5 / 57.0	53.7 / 71.8	25.5 / 30.0
A3	Who am I?	0	0.000	0.000	5.000	0.180	1.750	34.5 / 57.0	53.7 / 71.8	25.5 / 30.0
B1	Is this the old station …	3	0.056	−0.138	4.724	0.173	1.657	30.2 / 53.4	50.6 / 70.3	22.1 / 26.9
B2	Where am I, lost …	2	−0.216	0.003	5.006	0.206	1.816	39.1 / 62.5	54.4 / 73.1	29.2 / 34.0
B3	Who am I, when no one …	2	0.165	−0.484	4.032	0.160	1.434	23.7 / 39.5	45.4 / 62.3	17.2 / 19.9

The common-scale spans are 0.579, 0.606, 0.606, 0.588, 0.657 and 0.602 mm with all matched terms and 0.528, 0.606, 0.606, 0.574, 0.697 and 0.519 mm without function words; every min–max tile spans 1.2 mm. The smoothed transform changes the height by at most 0.042 mm with all matched terms and 0.050 mm without function words. At $N = 128$ the steepest point of every tile on the common scale lies on the edge of the domain. The field jumps there, since f is not an integer except at $(0, 0)$; the one-sided difference at the edge does not see the jump, but the level of the slope at a given N depends on where the window is cut. At $N = 1024$ the smoothed tiles change least: over the three sentences of text B, 51°–58° on the common scale, 68°–71° under min–max and 26°–31° with $T_{\gamma,\varepsilon}$.

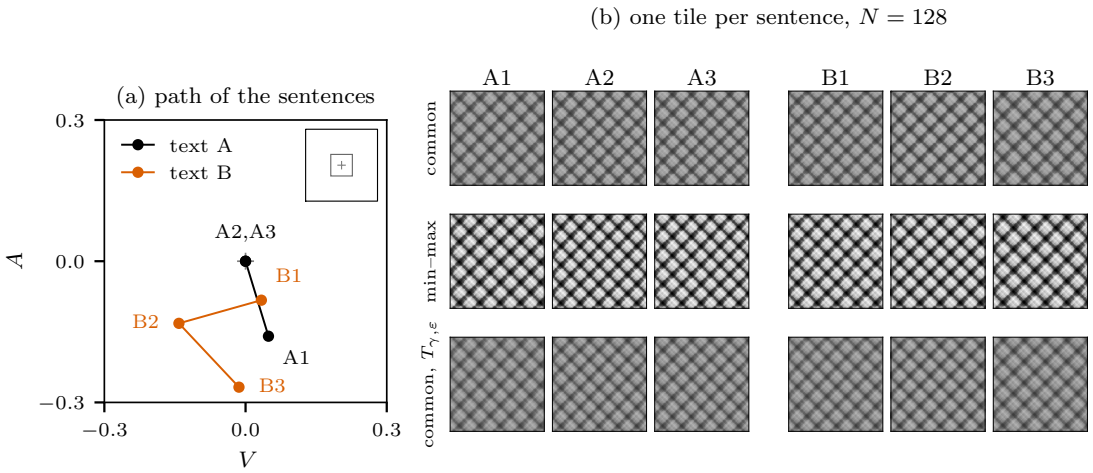


Figure S1: Texts A and B through the rule, all matched terms averaged.

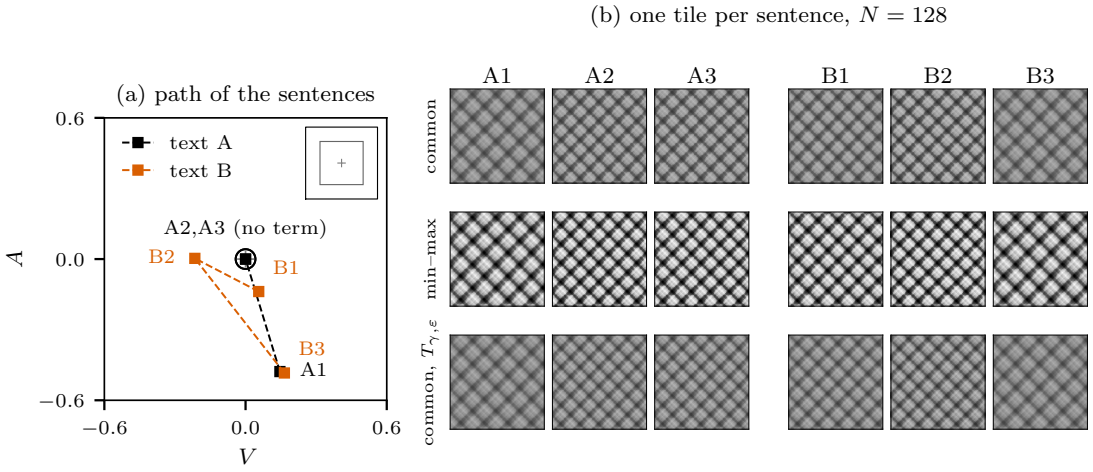


Figure S2: Texts A and B with function words removed before averaging.

S9 Reproduction

The repository is at `https://github.com/mikotomiura/affective-relief-audit` and is archived at Zenodo (doi:10.5281/zenodo.23681773). This supplement cites commits by their hashes in the working repository in which the computations were run; the published history was prepared from it by removing the LaTeX source of the manuscript and setting the author name, with the files and the dates unchanged, and `commit-map.tsv` in the repository maps each hash to its published commit. The repository holds the generator (`src/atact/`), the tests, the specification, the conditions files, the scripts and the results. Package versions are fixed by `uv.lock` (SHA-256 3c607b98…); every results file records the SHA-256 of its conditions file, its script and the specification, the package versions and the commit at which it was run. Each results file is regenerated by one command from the repository root:

Results	Command (`uv run python experiments/...`)
k2_gradient	`k2_gradient.py`
k4_centroid_2d	`k4_centroid_2d.py`
edge_grid_paired	`edge_grid_paired.py`
validation3_public	`validation3_public.py`
k5_power, k5_large_sample, k5_classification	`k5_classification.py` `--power`, `--large-sample`, `--frame`, then without options
table1_main	`table1_main.py`
variants_reported	`variants_reported.py`
text_path_example	`text_path_example.py`, then `text_path_figures.py`
observable_dof	`observable_dof.py`
k4_supplement, v3_reduction_supplement	`k4_supplement.py`, `v3_reduction_supplement.py`
cband_cutoff_sensitivity	`cband_cutoff_sensitivity.py`
Figures 1–4	`figures_main.py`

The NRC VAD Lexicon v2.1 is not redistributed, because its terms of use do not allow it; the text example needs the file `NRC-VAD-Lexicon-v2.1.txt` (SHA-256 42c71881…) from its authors. The measured data are read from their public records: the Surface-Topography Challenge (doi:10.5281/zenodo.15341939) and TactileDreamFusion on Hugging Face (revision 87c8b764bcb2ec1139de0c6435f45ac1d9f283d1).

S10 Literature searched

The claims of novelty in the main text (Section 1 and K4) rest on the following searches, made on 2026-10-24. They are not exhaustive.

The two-dimensional centroid. Web search, arXiv, HAL, institutional repositories, Crossref, Unpaywall and Semantic Scholar, for the decay of the Fourier transform of functions with singularities on curves and for spectral centroids or mean frequencies in relation to resolution and singularities. Read in the original (the sections that bear on the question): Brandolini, Hofmann and Iosevich (2003), Miao, Yan and Zhang (arXiv:2405.02531), Lagae et al. (2010), Gass (2025), lecture notes on homogeneous distributions and a preprint version of Candès and Donoho. The type of decay (an extra $|k|^{-1/2}$ from a curved singular line) is known. No source was found that derives an exponent of the centroid from it; the combination of 'spectral centroid' or 'mean frequency' with resolution and singularity returned no result. Not read (not freely available or refused to automated retrieval): Herz (1962), Randol (1969), and the general theorems of Hörmander, Stein (1993), Varchenko (1983) and Podkorytov on such decay.

The edge of the domain. Read in the original: Jacobs, Junge and Pastewka (2017), Moisan (2011), Persson et al. (2005); the documentation of two audio libraries (librosa, torchaudio) for the spectral centroid. The mechanism (the jump of the periodic extension adds a slowly decaying tail that windows suppress) and its effect on descriptors weighted towards high frequencies (RMS slope and curvature) are known. No source was found that states the growth of an amplitude-weighted centroid with the grid through leakage. Not read: the audio-descriptor reports of Peeters (2004) and Peeters et al. (2011).

Nearby work. Crossref, OpenAlex (ten queries, four restricted to preprints, 15–25 results each), Semantic Scholar and arXiv (eight queries), the authors' pages, and the titles of the Bridges proceedings 2012–2026. Read in the original: Naeem et al. (2025), TactStyle, Gao et al. (2026) and Elkharraz et al. (2014). None generates height maps from affective coordinates by an explicit formula, apart from the preprint read in this paper. Not searched individually: Research Square, SSRN, OSF and TechRxiv; the abstracts and texts of the Bridges proceedings.

Rules as works. Made on 2026-10-26 for Section 1 of the main text. Read in the original: Galanter (2003), including the passage that quotes LeWitt's *Paragraphs on Conceptual Art*, and Ippolito (2003) in *Permanence Through Change*. Not read: Nake (2010, refused to automated retrieval), van Saaze (2013, the repository link returned an error), LeWitt (1967) itself. A further search for the decay of Fourier transforms with curved singular sets found no freely available source beyond those listed above; Herz (1962) and Stein (1993) remain unread.

No paid full text was used.